Epistemic Collapse in Multi-Agent Systems

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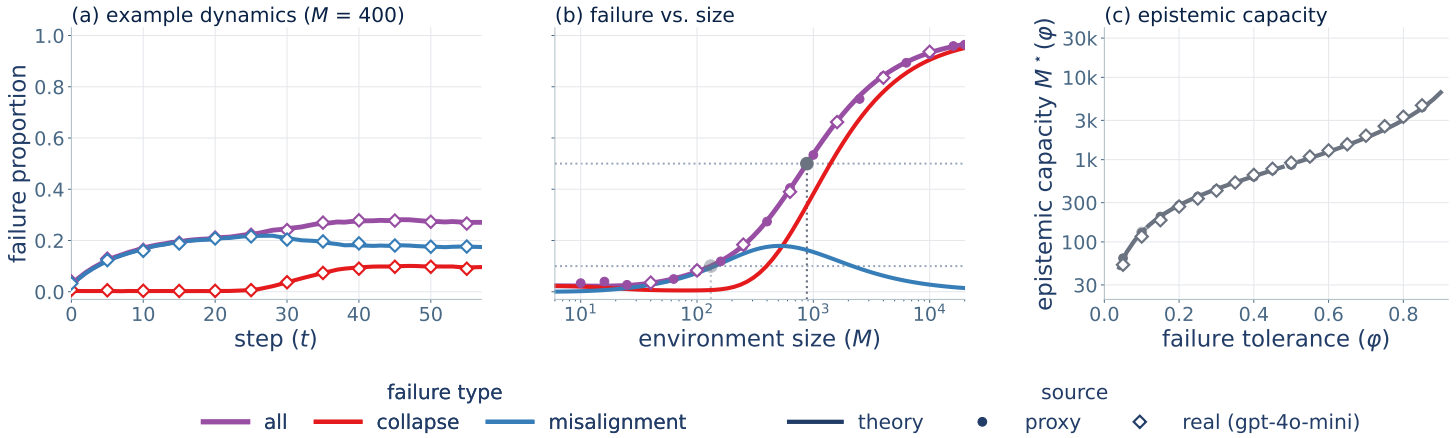


Figure 1: Epistemic collapse and epistemic capacity. Throughout the paper, color denotes the mitigation strategy and line style the information source (dashed = firsthand only, solid = uses secondhand information); violet denotes epistemic failure φ, the proportion of questions no agent answers correctly, split into collapse (dark) and misalignment (light). Every system in this figure is firsthand-only, so here solid and dashed instead distinguish fast ($c = 0.15$) and slow ($c = 0.05$) forgetting. *Left.* All agents start with knowledge of correct answers to all M questions in the environment; failure over time (mean ± 1 SEM over 20 seeds). *Center.* Steady-state failure ($t \geq 200$) vs. environment size; lines are theory, markers simulation. *Right.* Epistemic capacity $M^\star(\varphi)$, the largest environment sustainable with failure below φ, vs. environment speed (opacity: $\varphi = 0.1, 0.5$).

1 Motivation

Information is being produced and shared more than ever before. The landscape is not only larger but faster: short-form content, the 24-hour news cycle, and algorithmic social media feeds shorten the shelf life of what is current. For example, the near saturation of public discourse by the Epstein files, abruptly displaced by the conflict in the Middle East and the prospect of war with Iran, is a recent reminder that a story dominates collective attention only until the next one arrives — so the facts in circulation are revised more quickly than most of their consumers re-check them.

The cost of recirculating generated content is understood at the model level, where systems trained on their own outputs degrade [1]. It is far less known at the *system* level, where a system of agents that can observe and repeat information can drift away from a changing world's current truth even when every individual behaves reasonably. Modeling our information landscape as a multi-agent system in a dynamic environment, our goal is to investigate different strategies for maintaining a robust knowledge system and their sensitivity to various environmental and social parameters. Building on our earlier work on monitoring multi-agent systems (*Control Charts for Multi-agent Systems*, arXiv:2605.11135), we study these questions in a real network of LLM agents and in a mechanistically faithful proxy. Our investigation is organized around a failure mode we call *epistemic collapse*.

2 Setting

We model the information landscape as a multi-agent system evolving in discrete steps. The system has three components: an environment, a population of agents, and a communication topology. At each iteration, the information in the environment changes and each agent spends a per-step budget obtaining information — either via querying the environment directly or by asking a peer it is connected to. Particular instantiations (population size, rates, budgets) are reported with each figure.

Environment. The environment is defined as a set of M question and answer pairs and is the system's only source of truth. The environment is dynamic in that the correct answer to a question can change with each time step, with a query q's answer changing at rate $\lambda_q \sim \text{LogNormal}(\bar{\lambda}, \sigma_\lambda)$. We study the effect of size of the environment M and the average speed of the environment $\bar{\lambda}$ on epistemic collapse.

Agents. Our system is populated by agents each with a fixed context window of size k_{ctx}. Each agent is grounded in that they are instructed to respond to a question with "I don't know" if there is not enough evidence in their context to respond confidently. The k_{ctx} most semantically and temporally relevant knowledge, as measured by $\langle q, k \rangle \, e^{-c \cdot \Delta t_q}$, is put in an agent's context where Δt_q is the elapsed time since receiving the knowledge. An agent's knowledge is either *firsthand* — obtained by consulting the environment directly — or *secondhand* — received from a peer, and it retains this provenance as it moves: agents relay answers *verbatim*, so a secondhand copy is an exact copy of some earlier firsthand observation and never drifts by paraphrase. Whether secondhand knowledge counts as evidence enough to ground an answer — whether agents may repeat what they have merely heard — is a choice we call the *sharing policy*, and the subject of Sec. 3. We study the effect of the context size k_{ctx} and the speed of memory decay c on system dynamics.

Communication. The flow of information between the agents and the environment is constrained by a communication topology, in which the environment is a distinguished node. At a given iteration, each agent has a fixed bandwidth of B actions per step across its incident edges. It spends E on obtaining information from the environment and $K = B - E$ obtaining information from the peers it is connected to. An exchange on

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an environment edge (a query) returns the current truth. An exchange on a peer edge redistributes existing information. The reply (possibly an abstention) inserts into the asker's memory as received evidence. We study the effect of the average number of peer-to-peer edges in the communication topology, the total bandwidth B, budget allocation strategy E/K, and sharing policy (Sec. 3) on system dynamics.

Measurement. Every agent is asked every question in the environment at regular intervals. Each response is exactly one of *correct*, *stale* (an answer, but out of date), or an *abstention* / "I don't know", so P(correct) + P(stale) + P(IDK) = 1. We track these at two levels: *agent level* (the average member) and *system level* (whether *no* member can answer, and whether *some* member is correct).

A proxy system. The main object of our study is a system of LLM agents. Even with relatively dated models such as `gpt-4o-mini`, a single system evolution with a population of 10 agents run for 500 steps costs ∼ $5. At the scale required to establish trends across environmental, agentic, and topological parameters this cost can be prohibitively expensive. To mitigate it, we replace the LLM call with a model of its answering behavior. In particular, we estimate the probability of a response in each state by probing an LLM repeatedly and substitute the LLM in the system with a simple random sample from a multinomial distribution. The proposed proxy reproduces the statistics of the real system with median deviation 0.01 (Figure 2). Results below are produced with the proxy system unless otherwise stated.

2.1 Epistemic collapse

For a given q we define epistemic collapse as the system state in which every agent responds to q with "I don't know" (i.e., the knowledge of q has been forgotten by the whole population). Collapse is distinct from *staleness*, in which agents still answer but with an out-of-date answer, and is an intended property of a grounded agent. For an environment with M questions, we define χ-epistemic collapse as the state in which $0 \leq \chi \leq 1$ of the M questions are in epistemic collapse. We refer to χ as the collapse proportion. For a given system configuration — its population, per-agent budget and its allocation, memory, and sharing strategy — we define its *epistemic capacity* $M^\star(\chi)$ as the largest environment the system can sustain, in steady state, with fewer than a proportion χ of its questions in epistemic collapse. Equivalently, the curve $\chi(M)$ traced by the steady-state collapse proportion as the environment grows is the system's capacity curve, read at height χ (Fig. 1, right). To compare two configurations that differ in a single strategy s (e.g., sharing secondhand information, or books), we define the *epistemic efficiency* of s as $\Gamma_s(\chi) = M^\star_s(\chi)/M^\star_{\text{none}}(\chi)$: the factor by which s multiplies the environment the system can sustain before a proportion χ of it collapses.

Figure 1 shows collapse from the best possible start: every agent warm-started with current, firsthand knowledge of everything. Accuracy falls from ≈ 1 to a plateau set by memory decay (∼0.61 at $c = 0.05$, ∼0.14 at $c = 0.5$; left). Conditioned on being wrong, the fast-decay system fails purely by *forgetting* (100% abstention); the slow-decay system's errors are ∼90% forgetting and ∼10% *confident staleness* — rising to ∼25% in the early transient — with the stale share concentrated on the fastest-revising questions. Abstention is insensitive to how fast any fact revises, but highly sensitive to how many facts there are: steady-state P(IDK) climbs from 0.18 to 0.90 as the environment grows from 25 to 400 questions at fixed budget (right) — breadth, not speed, silences the population, because every new fact's insertions crowd the others out of retrievable memory. The same regimes appear in the real gpt-4o-mini factor grids (project artifacts): decay ($c : 0.05 \to 1.0$ takes agent IDK $0.35 \to 0.89$), source starvation (environment bandwidth $E : 6 \to 1$ takes system IDK $0 \to 0.29$), environment speed ($\bar{\lambda} : 0.001 \to 0.1$ takes system accuracy $0.93 \to 0.20$ without moving abstention).

The rest of this note studies the two natural *mitigations*, both forms of communication: (1) **sharing secondhand information** (mimesis) — letting agents repeat answers they received rather than observed; and (2) **books** — a shared artifact agents write to and read from. Both run on the validated platform of Sec. 2.

3 Mitigation 1: sharing secondhand information

The mimesis grid repeats the factor design of Sec. 2 with one change: received answers may ground an answer (still verbatim — agents repeat, they do not paraphrase). Figure **??** overlays the two grids at the agent level.

Mimesis raises the floor. At baseline, agent-level IDK drops $0.68 \to 0.26$ and agent accuracy rises $0.23 \to 0.48$. The communication factor switches from a null lever to a live one:

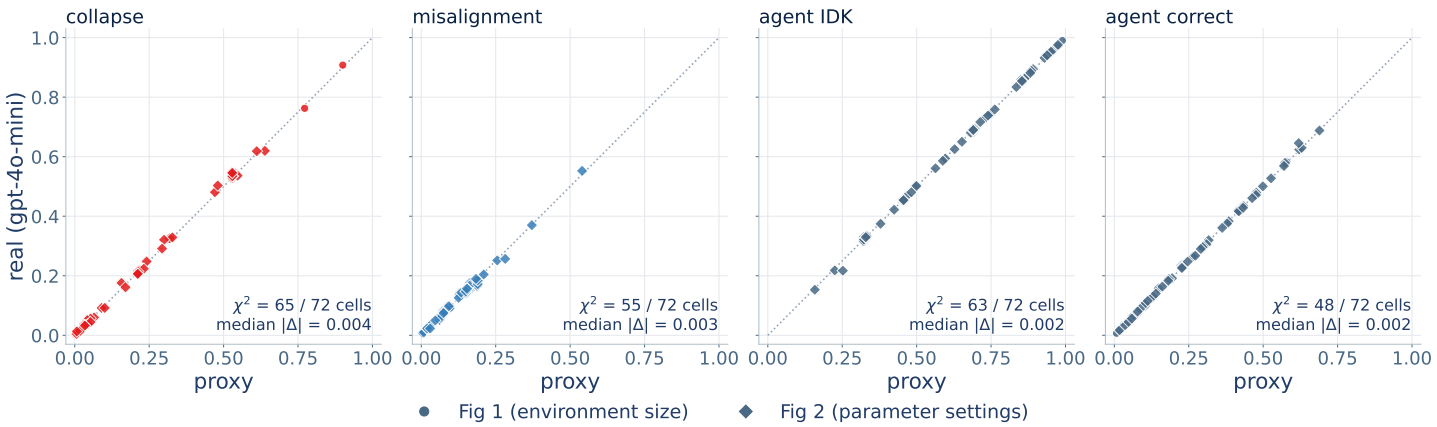


Figure 2: Validation of the proxy: the measured-gate simulation matches real gpt-4o-mini agent networks, with zero fitted parameters. Each panel varies one factor over three settings; each point is one setting × one metric, the median over proxy seeds (horizontal) against the median over real seeds (vertical). Color: no mitigation (gray) and sharing secondhand information (light blue); marker shape: metric. Median $|\Delta| = 0.01$ in every factor.

agent accuracy is 0.38 at degree 2 vs. 0.56 on the mesh, where without mimesis the same sweep is flat.

But not the ceiling. System-level accuracy does not improve — 0.72 (no mimesis) vs. 0.66 (mimesis) at baseline — because the system level asks whether *anyone* holds current evidence, and repetition redistributes evidence without creating it. Mimesis pays at the system level only where redistribution is the binding constraint: in the observation-starved cell ($E = 1$) system accuracy rises $0.46 \to 0.64$ and system IDK falls $0.29 \to 0.02$. The regime it cannot rescue is fast forgetting ($c = 1.0$: system IDK 0.33 vs. 0.31) — copies crowd out as fast as originals.

4 Mitigation 2: books

Mimesis is peer-to-peer. The second mitigation is a shared artifact: a *book* any agent can read for one action, fed only by deliberate, costly writes — an agent attempts a write with probability p, an attempt consumes its entire step, and a write commits after W attempts, merging the writer's current grounded knowledge (newest evidence wins). Reads ground answers like observations. We sweep the toy platform (10 seeds per point) in a scarce-observation regime where books have room to matter.

The book's value is a race between its write rate and the world's revision rate. The natural unit is the expected write rate $\omega = p/W$ (completed writes per agent per step). The book's accuracy gain over the no-books floor is large when $\bar{\lambda} \ll \omega$ (+0.48 at $\omega = 10/100$ steps, $\bar{\lambda} = 0.001$) and gone when $\bar{\lambda}/\omega \gtrsim 1$: every condition lands on the floor by $\bar{\lambda} = 0.3$ (Fig. **??**, left). Availability behaves completely differently: P(IDK) is set by ω (0.43/0.63/0.83 for $\omega = 10, 1, 0.2$ per 100 steps, vs. 0.93 without books) and is *flat in* $\bar{\lambda}$ (Fig. **??**, right). Staleness absorbs the entire loss: the book keeps everyone answering while the answers quietly expire — a fast world does not silence the library, it turns it into a confident source of expired facts.

ω is the sufficient statistic. Holding p fixed and varying W (constant budget tax) reproduces the ordering, confirming W is a pure freshness dial with diminishing returns once the book turns over faster than the world. Conversely, holding ω fixed while varying p by 4–10× moves the curves by at most 0.06 on either metric, against between-ω gaps up to 0.22. The residual is monotone in p with the sign a budget tax predicts (high-p writers lose more steps to attempts and deposit slightly staler content, $\approx$ 0.04–0.06 accuracy at slow speeds). For design purposes: quote a book policy by its expected writes; the attempt probability is a second-order tax.

Takeaway and next steps. Communication — secondhand sharing or shared artifacts — reliably buys *availability*, and buys *accuracy* only while the underlying evidence is younger than the world's revision clock: mimesis is capped by what the population has observed, and a book by how often anyone can afford to write. As the environment speeds up, the failure mode is not silence but confident staleness, which availability metrics cannot see. Next: (1) "dated editions" — books exposing their write time so readers can abstain on stale entries, trading availability back for accuracy; (2) strategy-sensitivity sweeps on the validated platform; (3) the monitoring question: does a control chart on the probe panel see confident staleness coming, given it presents as drift, not shift?

References

[1] Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. AI models collapse when trained on recursively generated data. *Nature*, 631:755–759, 2024.